Facultad de Defensa Nacional

Carrera Lic. En Ciberdefensa

Análisis Matemático I

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Carga horaria: 64

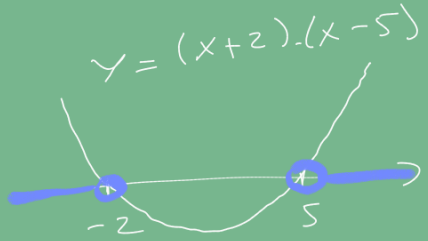
FUNCIÓN: $f : A \longrightarrow B$

A: DOMINIO — B: CODOMINIO

Ej: $f(x) = \dfrac{x^3}{\sqrt{(x+2)(x-5)}}$ HALLAR $\mathrm{Dom}(f)$.

DOMINIO NATURAL.

$x \in \mathrm{Dom}(f) \iff (x+2).(x-5) > 0$



$\mathrm{Dom}(f) = (-\infty, -2) \cup (5, +\infty)$

$f : (-\infty, -2) \cup (5, +\infty) \longrightarrow \mathbb{R}$

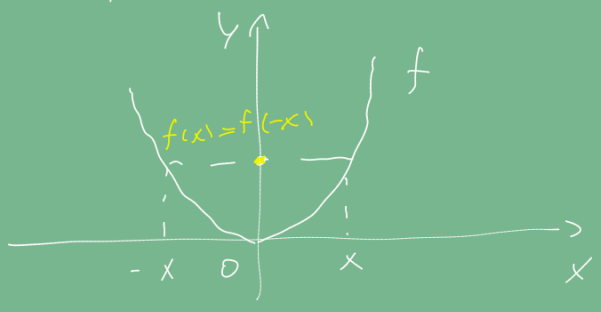
$f(x) = \dfrac{x^3}{\sqrt{(x+2)(x-5)}}$

FUNCIONES PARES:

UNA FUNCIÓN f ES PAR SI

$(\forall x \in \mathrm{Dom}(f)) \quad f(x) = -f(x)$.

Ej: $f : \mathbb{R} \longrightarrow \mathbb{R}$, $f(x) = x^2$.



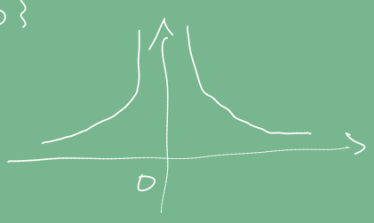
Ej
$g(x) = x^4$

$h(x) = x^6$

$(\forall x) \quad f(-x) = (-x)^2 = (-1)^2 . x^2 = 1 x^2 = x^2 = f(x)$.

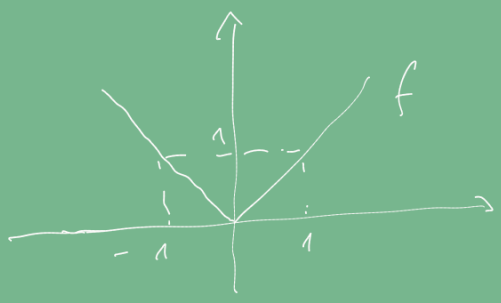
Ej: $f : \mathbb{R} - \{0\} \longrightarrow \mathbb{R} - \{0\}$

$f(x) = \dfrac{1}{x^2}$



Ej.: $f(x) = |x|$

$f(-x) = |-x| = |x| = f(x)$

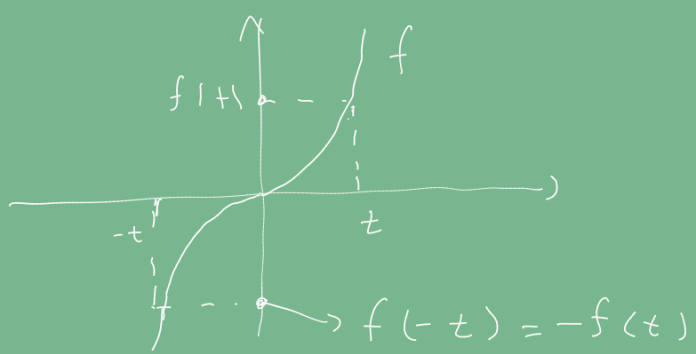


FUNCIONES IMPARES:

f ES IMPAR si $(\forall x)\ f(-x) = -f(x)$.

Ej: $f(x) = x^3$, $f(-x) = (-x)^3 = (-1 \cdot x)^3 =$

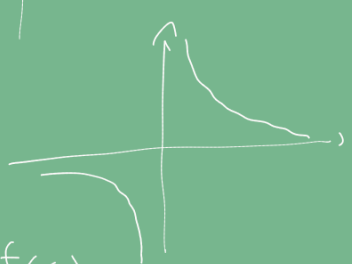
$= (-1)^3 \cdot x^3 = -1 \cdot x^3 = -x^3 = -f(x)$



Ej: $f(x) = x$



Ej: $f(x) = \frac{1}{x}$



$f(-x) = \frac{1}{-x} = -\frac{1}{x} = -f(x)$

OBS.: SI f ES IMPAR Y f ESTÁ DEFINIDA EN $x=0$, ENTONCES $f(0)=0$.

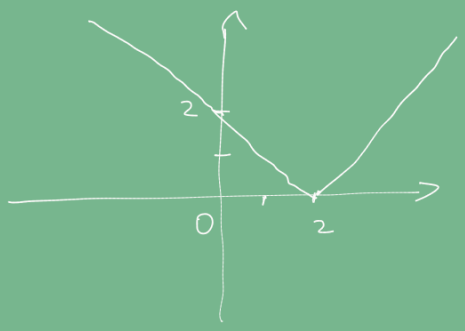
$f(0) = f(-0) = -f(0) \Rightarrow f(0) = -f(0) \Rightarrow$

$\Rightarrow 2f(0) = 0 \Rightarrow \underline{f(0) = 0}$

Ej.: $f(x) = |x-2|$

$g(x) = |x|$

$f(x) = g(x-2)$



$f(1) = |1-2| = |-1| = 1$

$f(-1) = |-1-2| = |-3| = 3$

$f(-1) \neq f(1) \Rightarrow$ f NO ES PAR.

$f(-1) \neq -f(1) \Rightarrow$ f NO ES IMPAR.

EL MÓDULO ES PAR

Ej: $f(x) = |-5x| = |5x|$

$f(-x) = |-5 \cdot (-x)| = |5x| = f(x), \forall x \in \mathbb{R}.$

$\therefore$ f ES PAR

OBS: $|x|$ ES PAR.

$$|x| = \begin{cases} x & \text{si } x \geq 0 \\ -x & \text{si } x < 0 \end{cases}$$

$$|-x| = \begin{cases} -x & \text{si } -x \geq 0 \\ -(-x) & \text{si } -x < 0 \end{cases} = \begin{cases} -x & \text{si } x \leq 0 \\ x & \text{si } x > 0 \end{cases} =$$

$$= \begin{cases} -x & \text{si } x < 0 \\ x & \text{si } x \geq 0 \end{cases} = |x|.$$

OTRA FORMA: $|x| = \sqrt{x^2}$

$|-x| = \sqrt{(-x)^2} = \sqrt{x^2} = |x|$